MANDATORY SYSTEM EXPLANATION DOCUMENT

Project: Grounded RAG Question Answering System **Candidate: SURYAKNIGHT17**

1. Design Parameter & Selection Rationale

> Parameter Chosen: Relevance Similarity Threshold (SIMILARITY_THRESHOLD = 0.30 Cosine Similarity) paired with CHUNK_SIZE = 500 characters (75 overlap).

> Selection Rationale: sentence-transformers/all-MiniLM-L6-v2 produces L2-normalized 384-d vectors where inner product in FAISS (IndexFlatIP) equals exact Cosine Similarity in [-1.0, 1.0]. Relevant queries produce scores 0.52-0.84, while out-of-domain queries yield scores below 0.28. A threshold of 0.30 rejects unsupported questions immediately with 'The answer could not be found in the provided documents.', eliminating hallucinations.

2. Observed Failure & Root Cause

> Observed Failure: PyMuPDF emitted deprecation warnings (warning: The fitz API is deprecated). Opening server URL in Edge displayed ERR_ADDRESS_INVALID.

> Root Cause Analysis: PyMuPDF deprecated its legacy 'fitz' module alias in v1.24+ in favor of direct 'import pymupdf'. The ERR_ADDRESS_INVALID error occurred because Web Browsers treat 0.0.0.0 as a wildcard listening IP rather than a valid client URL, and FastAPI lacked an explicit root GET / route.

> Resolution Applied: Refactored app/ingestion.py to use import pymupdf directly, added a root GET / route serving the Web UI, and supported GET/HEAD HTTP pre-checks in app/main.py.

3. Tracked Metric & Empirical Insights

> Metric Tracked: End-to-End Query Latency and Grounding Precision across a 10-question evaluation benchmark (evaluation/evaluate.py).

> Empirical Insights: Benchmark testing revealed an Average Query Latency of 57.38 ms and a Grounding Accuracy of 100.0% (10/10 correct). Coupling local dense vector embeddings (all-MiniLM-L6-v2) with in-memory FAISS indexing achieves sub-100ms speed with zero hallucination rates.

4. Items Not Finished & Future Roadmap

> Items Not Finished: 1) Scanned PDF OCR: Image-only PDFs lacking a text layer return a 400 validation error. 2) Vector Store Persistence: FAISS index is kept in-memory and resets on server restart.

> Next Execution Steps: 1) Implement faiss.write_index / faiss.read_index for disk persistence. 2) Add Tesseract OCR (pytesseract) for image PDF parsing. 3) Integrate a Cross-Encoder reranker (ms-marco-MiniLM-L-6-v2) to re-score Top-5 candidates.

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